

PET HEALTH AND SAFETY

Chocolate (methylxanthine) toxicosis in dogs: dose thresholds, decontamination and treatment

Veterinary version: clinical detail and references. Switch to the owner version for plain-language steps.

DH By Dr Duncan Houston · Emergency vet · 13 min read

Chocolate toxicosis is a dose-per-kilogram problem with a long toxicokinetic tail, so the calculated methylxanthine dose and the signs in front of you decide the plan together. This version gives the dose thresholds, the kinetics behind the monitoring window, the decontamination decision by time since ingestion, and cited doses for complications above 40 mg/kg ¹². Of 156 dogs that ate chocolate, 44 developed signs and 43 of those survived ³.

01

Clinical summary

Mild signs begin at about 20 mg/kg of total methylxanthines, cardiotoxic effects at 40 to 50 mg/kg and seizures at 60 mg/kg and above ¹². Signs start 2 to 12 hours after ingestion, plasma theobromine peaks at about 10 hours, and the theobromine half-life is 17.5 hours ¹². Admit any dog at or above 40 mg/kg, or with tachycardia, tremors or seizures at any calculated dose, for ECG monitoring and IV fluids ¹. The 20 to 40 mg/kg band gets same-day emesis when clinically normal (effective decontamination within 2 to 4 hours carries a good prognosis ²; chocolate fat slows gastric emptying), then observation; under 20 mg/kg with no signs goes home with a 24 hour watch derived from the half-life and a written return trigger ¹. Mortality among symptomatic dogs was below 3% ³.

02

Triage tiers

Calculated dose (mg/kg)	Expected signs	Disposition
Under 20, or white chocolate only (an insignificant methylxanthine source) ¹²	None from methylxanthines; GI upset from fat and sugar, pancreatitis in susceptible dogs ¹	Monitor at home 24 hours ¹ ; return for repeated vomiting, restlessness or tachycardia
20 to 40 ¹	Vomiting, diarrhea, polydipsia, restlessness ¹	Same day: emesis if clinically normal, best inside 2 to 4 hours and later when the dose justifies it ² , then observe
40 to 60 ¹	Sinus tachycardia, arrhythmia, hyperactivity, tremors, hyperthermia, hypertension ¹	Emergency now: admit, continuous ECG, IV fluids
60 and above ¹	Seizures, ataxia, weakness, coma ¹	Emergency now: admit, seizure and temperature control, continuous ECG
Unbounded: unmeasured amount, dark or baking chocolate in a dog under 10 kg, any puppy	Any of the above	Emergency now: treat as 40 mg/kg or more until measured

03

Pathophysiology and toxicokinetics

Theobromine and caffeine competitively antagonize adenosine receptors, which produces CNS stimulation, diuresis, vasoconstriction and tachycardia ¹. They raise intracellular calcium in striated muscle (the mechanism behind rigidity and tremors), inhibit phosphodiesterase, may raise circulating epinephrine and norepinephrine, and compete at CNS benzodiazepine receptors ¹.

Caffeine is absorbed rapidly; theobromine is absorbed about 10 times more slowly and peaks in plasma at about 10 hours ². After hepatic metabolism the methylxanthines undergo enterohepatic recycling and are excreted in urine as metabolites and unchanged parent compound ¹. The theobromine half-life is 17.5 hours and the caffeine half-life 4.5 hours ¹. Methylxanthines and their metabolites are reabsorbed across the bladder wall, the basis for urinary catheterization in the recumbent patient ¹. Signs are reported from 2 to 4 hours after ingestion ² and within 6 to 12 hours ¹. The theobromine LD50 in dogs is reported as 250 to 500 mg/kg ¹; life-threatening arrhythmias and CNS dysfunction are the fatal mechanisms, and the fat load can trigger pancreatitis ¹.

Product	Total methylxanthines ¹
Dry cocoa powder	28.5 mg/g (808 mg/oz)
Unsweetened baking chocolate	16 mg/g (454 mg/oz)
Cocoa bean hulls (garden mulch)	9.1 mg/g (258 mg/oz)
Semisweet or dark chocolate	5.4 to 5.7 mg/g (153 to 162 mg/oz)
Milk chocolate	2.3 mg/g (65 mg/oz)
White chocolate	Insignificant ¹²

Other stimulant toxicoses present the same way: caffeine tablets at about 100 mg each ¹, amphetamines, pseudoephedrine, cocaine and guarana. Take the co-ingestant history at the same time: xylitol in sugar-free chocolate (see [Xylitol poisoning in dogs: signs and what to do](https://askavet.com/blogs/news/xylitol-poisoning-in-dogs-2025-a-deadly-risk-hiding-in-gum-peanut-butter-toothpaste) (<https://askavet.com/blogs/news/xylitol-poisoning-in-dogs-2025-a-deadly-risk-hiding-in-gum-peanut-butter-toothpaste>)), chocolate-coated raisins (see [Grapes and raisins poisoning in dogs](https://askavet.com/blogs/news/the-wrath-of-grapes-2025-why-raisins-grapes-are-fatal-to-dogs) (<https://askavet.com/blogs/news/the-wrath-of-grapes-2025-why-raisins-grapes-are-fatal-to-dogs>)), macadamia nuts, coffee beans and foil.

04

Dose calculation and thresholds

Dose in mg/kg = grams ingested x mg/g for the product, divided by body weight in kg. Use the upper figure of any content range and the owner's upper estimate. Worked example: an 18 kg dog eats 30 g of dark chocolate: $30 \times 5.7 = 171$ mg, and $171 / 18 = 9.5$ mg/kg, below the 20 mg/kg threshold ¹.

Thresholds: mild signs at 20 mg/kg, cardiotoxic effects at 40 to 50 mg/kg, seizures at 60 mg/kg and above ¹². Of 44 dogs with signs, 28 had eaten dark or bitter chocolate ³. Treat the lines as bands: susceptibility varies with the individual, an empty stomach and the product ¹. Among 185 UK dogs whose dose could be calculated, 75 (41%) had eaten less than 20 mg/kg ⁴.

The table gives the grams that reach 20 mg/kg at the content figures above ¹; double for the 40 mg/kg cardiac band, triple for the 60 mg/kg seizure band ¹.

Body weight	Milk chocolate (2.3 mg/g)	Dark or semisweet (5.7 mg/g)	Baking chocolate (16 mg/g)
5 kg	43 g	17 g	6 g
10 kg	87 g	35 g	12 g
20 kg	173 g	70 g	25 g
30 kg	260 g	105 g	37 g

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Clinical signs by severity

Onset is 2 to 12 hours ¹². Initial signs are nausea, vomiting, diarrhea, polydipsia and polyuria; progression runs to restlessness, hyperactivity, tachypnea, ataxia, tremors, seizures, weakness and coma, with arrhythmia, hypertension and hyperthermia ¹. Death follows arrhythmia or CNS dysfunction ¹.

Among 44 dogs with signs, the reasons for presentation were agitation (33), tremor (22), vomiting (21), panting (11), polyuria and polydipsia (7) and diarrhea (2); examination found sinus tachycardia in 28, tachypnea or panting in 14, hyperthermia in 10 and dehydration in 7 ³. Of the 34 with abnormal results, 23 had hyperlactatemia, 16 hypokalemia, 16 mild hyperglycemia and 14 mild ALT and AST elevation ³. At the extreme, a 2.2 kg puppy that ate about 176 mg/kg from 200 g of homemade chocolate cake had dyspnea at about 4 hours, a heart rate of 170 bpm and a respiratory rate of 80 at 5 hours, and non-cardiogenic pulmonary edema on radiographs ⁵.

Red flags:

- **Persistent sinus tachycardia or premature ventricular contractions.** Cardiotoxic band 40 to 50 mg/kg ¹; 28 of 44 symptomatic dogs ³.
- **Tremors or rigidity.** One step before seizures ¹; 22 of 44 symptomatic dogs ³.
- **Seizure.** Expected from 60 mg/kg ¹; recorded 4.5 hours after ingestion at 88.3 mg/kg ¹⁴.
- **Hyperthermia.** 10 of 44 symptomatic dogs ³; listed among the terminal signs ¹.
- **Tachypnea, cyanosis or crackles.** Pulmonary edema at 4 hours in the 2.2 kg puppy ⁵.
- **Bradycardia or coma.** Late-stage signs ¹.

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Telephone triage advice

Collect body weight in kg, the product, the grams missing at the upper estimate and the time of ingestion, and calculate on the line. Come now for 20 mg/kg or above ¹, for any sign already present, for dark or baking chocolate in a dog under 10 kg, for any puppy and for any unmeasurable amount.

No home emesis. Prehospital emesis has no evidence of benefit in toxin ingestion and carries risk, so routine emesis at home is advised against and the animal is presented to a veterinary facility as soon as possible ⁶. Oral 3% hydrogen peroxide produced gastric mucosal lesions in every dog given it, worse at 24 hours, with esophagitis in some ⁸; salt causes hypernatremia. The exception is a remote location where a veterinarian or a poison center advises field emesis because no clinic is reachable ⁶. Leave water available. Bring the wrapper, the last recorded weight, the time of ingestion and any vomitus in a bag.

07

Decontamination, treatment and monitoring

Emesis. Decontaminate the dog that presents before signs develop ¹; effective decontamination within 2 to 4 hours of ingestion carries a good prognosis ², and chocolate fat slows gastric emptying, so a later presentation is judged on the dog's state, the size of the ingestion and what comes up, not the clock. Among 375 UK dogs, 26% presented within 1 hour and 56% within 6 hours ⁴. Do not induce in a dog that is tremoring, seizing, obtunded or already vomiting. Options: apomorphine 0.03 mg/kg IV produced emesis within 10 minutes in 22 of 25 dogs and 0.06 mg/kg intranasally in 18 of 25, against 14 of 25 for ropinirole eye drops, 6 of 25 for the SC route and 4 of 25 for the compounded injectable given transconjunctivally ⁹. Ropinirole 3.75 mg/m² to the eye produced vomiting in 95% of 100 dogs within 30 minutes, median 10 minutes ¹⁰; in an emergency-room comparison its first-dose success was 81% against 99% for IV apomorphine, with a median time to emesis of 8.6 against 1.6 minutes and rescue antiemetics needed in 37% against 0% ¹¹. Apomorphine and 3% hydrogen peroxide were similarly effective (94% and 90%) with mild adverse effects in 14% ⁷, but hydrogen peroxide injures the gastric mucosa ⁸, so it has no place in the clinic either.

Activated charcoal. The cited dose is 1 to 4 g/kg PO, repeated every 3 hours for up to 72 hours in symptomatic animals because of enterohepatic recycling ¹; multiple doses are the cited mainstay with supportive care ². Hypernatremia is the risk argued against it. In 96 dogs given a single dose with or without sorbitol, no dog developed hypernatremia (sodium above 155 mEq/L) and median sodium fell rather than rose at every checkpoint to 48 hours ¹². In 97 dogs given 2 to 6 doses, 95 of them (98%) on IV fluids, no dog exceeded 155 mEq/L and all survived ¹³. The counter-case: a 7-year-old Corgi at 88.3 mg/kg, given emesis, maropitant and charcoal with sorbitol before transfer, seized at 4.5 hours, reached a sodium of 174 mmol/L with neurologic signs and azotemia, needed ventilation, and went home at 57 hours with vestibular signs that resolved within 10 months ¹⁴. The protocol that follows: charcoal for the high-dose patient; IV fluids every time; sodium at presentation and at 6 to 12, 12 to 24 and

24 to 48 hours, the checkpoints at which the two series measured it ¹²¹³; repeat doses only in the monitored inpatient whose dose justifies them, with vomiting controlled (metoclopramide 0.2 to 0.4 mg/kg SC or IM) ¹.

Vet tool: emergency and critical care

Sodium Correction Calculator

Free water deficit and a safe correction rate for the post-charcoal sodium that came back high.

Open the calculator (<https://askavet.com/pages/sodium-correction-calculator>)

IV fluids. Fluid diuresis supports cardiovascular function and hastens urinary excretion of methylxanthines ¹; forced diuresis was part of the treatment in the 43 survivors ³; rate and type per your formulary. Hypokalemia affected 16 of the 34 dogs with abnormal results, so check potassium at admission ³. Erythromycin and corticosteroids are avoided because they interfere with methylxanthine excretion ¹.

Antiemetics. After emesis is complete and never before it: metoclopramide 0.2 to 0.4 mg/kg SC or IM ¹, or maropitant per your formulary.

Tachyarrhythmia. Correct hypovolemia, hyperthermia and anxiety first. For ventricular tachyarrhythmias, lidocaine 1 to 2 mg/kg IV to effect, then a constant-rate infusion per your formulary; lidocaine is not used in cats ¹. If lidocaine fails, metoprolol or propranolol 0.02 to 0.06 mg/kg slow IV, no faster than 1 mg per 2 minutes, repeated and increased to effect; metoprolol is preferred because propranolol can slow renal excretion of methylxanthines ¹. Esmolol was the beta blocker used in the 156-dog treatment protocol ³. For bradycardia, atropine 0.01 to 0.02 mg/kg IM or SC ¹. The puppy with pulmonary edema went home on day 6 after IV fluids, oxygen, furosemide 2 mg/kg IV three times daily, a bronchodilator and an antiemetic ⁵.

Vet tool: emergency and critical care

Emergency Drug Calculator

Crash-cart doses and draw-up volumes by body weight.

Open the calculator (<https://askavet.com/pages/emergency-drug-calculator>)

Tremors and seizures. Diazepam 0.5 to 2 mg/kg slow IV, or barbiturates ¹; methocarbamol for tremors, and levetiracetam or general anesthesia for refractory seizures, per your formulary. Because methylxanthines compete at benzodiazepine receptors ¹, a partial response to the first diazepam dose is a reason to escalate rather than repeat.

Urinary catheter. Place one in the recumbent patient, because methylxanthines and their metabolites are reabsorbed across the bladder wall ¹; walk the ambulatory patient to void at frequent intervals.

Monitoring and duration. Continuous ECG in the cardiac band and above, temperature, acid-base and electrolytes ¹, and sodium on the charcoal schedule ¹²¹³. Plan from the 17.5 hour half-life ¹: the cardiac-band dog stays on the ECG through at least 1 half-life, and the dog that seized stays until sign-free for a full half-life. The hypernatremic Corgi went home at 57 hours ¹⁴ and the puppy with pulmonary edema on day 6 ⁵. With that care 43 of 44 symptomatic dogs survived ³.

08

Common management errors

1. Calculating with the lower content figure or the owner's low estimate; dark chocolate runs to 5.7 mg/g ¹.
2. Refusing emesis at hour 3 because a fixed window has passed; the good-prognosis window is 2 to 4 hours ² and chocolate fat slows gastric emptying, so the decision rests on the dog and the dose.
3. Charcoal for the low-dose, asymptomatic dog; 41% of dose-known UK cases were below 20 mg/kg, and 34 of the 75 non-toxic cases were still given apomorphine ⁴.
4. Charcoal without fluids and without a sodium check ¹²¹⁴.
5. Antiemetic before emesis.
6. A beta blocker for a sinus tachycardia that hypovolemia, hyperthermia or fear explains, and no ECG in a dog above 40 mg/kg ¹.
7. Discharge at 6 hours because the dog looks normal; onset runs to 12 hours and the half-life is 17.5 hours ¹.
8. A full bladder in a recumbent patient ¹.
9. Erythromycin or a corticosteroid during recovery ¹.

09

Prevention counseling

Chocolate goes above counter height in a closed cupboard, and baking chocolate and cocoa powder are treated like medication: 6 g of baking chocolate brings a 5 kg dog to the 20 mg/kg line ¹. Cocoa mulch at 9.1 mg/g goes on the list for any client with a garden ¹.

Chocolate exposure in 375 UK dogs was 4.7 times more likely at Christmas and 2 times more likely at Easter than at other times of year, and dogs under 4 years were the most exposed ⁴. A trained "leave it" cue and floor-level nose work give a bored dog a job; the owner version expands on both.

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Clinical FAQs

Is emesis still worth inducing 4 hours after ingestion?

Yes for the clinically normal dog with a meaningful calculated dose. The good-prognosis window for decontamination is 2 to 4 hours ², chocolate fat slows gastric emptying, and the yield is judged by what comes up.

Should activated charcoal be repeated for chocolate?

Enterohepatic recycling is the argument for it, and the cited schedule is every 3 hours for up to 72 hours in symptomatic animals ¹; in 97 dogs given 2 to 6 doses on fluids no dog became hypernatremic ¹³. Repeat only in the hospitalized high-dose patient on fluids, with sodium checked at 6 to 12, 12 to 24 and 24 to 48 hours ¹²¹³, because a single charcoal dose with sorbitol preceded a sodium of 174 mmol/L in the Corgi ¹⁴.

Do cats need the same protocol?

Chocolate is toxic to every species ¹, so run the same calculation and downstream protocol, without lidocaine ¹. Cats are harder to make vomit: among 316 Australian veterinarians surveyed, emesis succeeded in 58% of cats against 99% of dogs, and most use IM dexmedetomidine or medetomidine for cats ¹⁶. Dexmedetomidine IM or IV produced emesis in 13 of 16 cats at a median dose of 7 mcg/kg (range 0.96 to 10), where hydrogen peroxide produced none in 3 ¹⁵.

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Bottom line

Calculate with the upper content figure and the upper estimate of grams, decontaminate the clinically normal dog, and keep every dog at or above 40 mg/kg on an ECG through at least one 17.5 hour half-life ¹. My opinion is that the sodium check after charcoal matters more than the second charcoal dose.

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Owners discharged with a low-dose dog can log the ingestion time, the calculated dose and every vomit or heart-rate check in [ASK A VET™](https://askavet.com) (<https://askavet.com>), which tracks a patient's signs hour by hour and puts the owner in chat with a certified vet when a return trigger is met.